

Statistical Inference for Variance (Chi-Square Test)

The **Chi-square test for variance** is used to determine whether the variance of a population differs from a specified value. It is especially useful when we want to check consistency, reliability, or variability in a process.

Unlike t-tests (which focus on means), this test focuses on **spread (variance)** and assumes that the data comes from a **normally distributed population**.

when to Use Chi-Square Test for Variance

Use this test when:

- You want to test a **single population variance**
- The population is **normally distributed**
- You are comparing:
 - Sample variance vs claimed variance
 - Process variability vs standard value

Point Estimate

$$\sigma^2 = s^2$$

Where $s^2 = \frac{\sum(X-\bar{X})^2}{n-1}$, *unbiased sample variance*

Confidence Interval for Variance

$$\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2}, n-1}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}}$$

Elementary Statistics by Bluman Exercise 7.4

Question# 12. **Home Ownership Rates** The percentage rates of home ownership for 8 randomly selected states are listed below. **Estimate** the population variance and standard deviation for the percentage rate of home ownership with **99% confidence**.

66.0, 75.8, 70.9, 73.9, 63.4, 68.5, 73.3 & 65.9

$$n = 8, s^2 = 19.7870, \frac{\alpha}{2} = 0.5\% = 0.005, 1 - \frac{\alpha}{2} = 0.995,$$

$$\chi^2_{\frac{\alpha}{2}, n-1} = \chi^2_{0.005, 7} = 20.278, \chi^2_{1-\frac{\alpha}{2}, n-1} = \chi^2_{0.995, 7} = 0.989$$

For Variance:

$$\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2}, n-1}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}} \rightarrow \frac{7*19.7870}{20.278} < \sigma^2 < \frac{7*19.7870}{0.989}$$

6.8305 to 140.0495

We are 99% confident that pop variance lies between 6.8305 to 140.0495

For Standard Deviation:

$$\sqrt{\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2}, n-1}}} < \sigma < \sqrt{\frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}}}$$

2.6135 to 11.83425

We are 99% confident that pop s.d lies between 2.6135 to 11.83425

Hypothesis Testing of Variance

There are 6 basic steps:

1. State the claim & its opposite
2. Level of significance (α)
3. Test statistic
4. Critical region
5. Calculation
6. Conclusion

Roadmap of Hypothesis testing for a single variance

Hypothesis	Test Statistics	Critical region
$H_1: \sigma^2 > \sigma_0^2$	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$	$\chi^2 > \chi^2_{\alpha, n-1}$
$H_1: \sigma^2 < \sigma_0^2$		$\chi^2 < \chi^2_{1-\alpha, n-1}$
$H_1: \sigma^2 \neq \sigma_0^2$		$\chi^2 > \chi^2_{\frac{\alpha}{2}, n-1}$ or $\chi^2 < \chi^2_{1-\frac{\alpha}{2}, n-1}$

Elementary Statistics By Bluman Exercise:8.5

Question#14. Heights of Volcanoes A sample of heights (in feet) of active volcanoes in North America, outside of Alaska, is listed below. Is there sufficient evidence that the standard deviation in heights of volcanoes outside Alaska is less than the standard deviation in heights of Alaskan volcanoes, which is 2385.9 feet? Use $\alpha=0.05$.

10777, 8159, 11240, 10456, 14163 & 8363

Step#1 Ho: $\sigma^2 \geq 5692518.81$

H₁: $\sigma^2 < 2385.9^2 \rightarrow \sigma^2 < 5692518.81$

Step#2 Level of significance: $\alpha = 0.05$

Step#3 Test Statistics

$$\chi^2 = \frac{(n-1)s^2}{\sigma_o^2}$$

Step#4

Critical region

$$\chi^2 < \chi_{1-\alpha, n-1}^2$$

$$\chi^2 < \chi_{0.95, 5}^2$$

$$\chi^2 < 1.145$$

If critical region satisfies reject Ho

Step#5 Calculation:

10777, 8159, 11240, 10456, 14163 & 8363

$$n = 6, s^2 = 4817344.667, \sigma_o^2 = 5692518.81$$

$$\chi^2 = \frac{(n-1)s^2}{\sigma_o^2} = \frac{5 * 4817344.667}{5692518.81} = 4.23129$$

4.23129 < 1.145 $\rightarrow$ doesn't satisfied

Step#6 Conclusion: As Cr Doesn't satisfied therefore there is no evidence to reject Ho at 0.05 level of significance we conclude that standard deviation is greater & equal to 2385.9

Question# 12. **Interstate Speeds** It has been reported that the standard deviation of the speeds of drivers on Interstate 75 near Findlay, Ohio, is 8 miles per hour for all vehicles. A driver feels from experience that this is very low. A survey is conducted, and for 50 drivers the standard deviation is 10.5 miles per hour. At $\alpha = 0.05$, is the driver correct?

Step#1 Ho: $\sigma^2 \leq 64$

H₁: $\sigma^2 > 64$

Step#2 Level of significance: $\alpha = 0.05$

Step#3 Test Statistics

$$\chi^2 = \frac{(n-1)s^2}{\sigma_o^2}$$

Step#4

Critical region

$$\chi^2 > \chi_{\alpha, n-1}^2$$

$$\chi^2 > \chi_{0.05, 49}^2$$

$$\chi^2 > \chi_{0.05, 49}^2$$

$$\chi^2 > 67.505$$

If critical region satisfies reject Ho

Step#5 Calculation:

A survey is conducted, and for 50 drivers the standard deviation is 10.5 miles per hour.

$$n = 50, s^2 = 10.5^2 = 110.25, \sigma_o^2 = 64$$

$$\chi^2 = \frac{49 * 110.25}{64} = 84.4102$$

84.4102 > 67.505 ➔ Satisfied

Step#6 Conclusion: AS CR satisfied therefore there is evidence to reject Ho at 0.05 level of significance we conclude that sd is greater than 8.

Question# 6. Carbohydrates in Fast Foods The number of carbohydrates found in a random sample of fast-food entrees is listed below. Is there sufficient evidence to conclude that the variance differs from 100? Use the 0.05 level of significance.

53, 46, 39, 39, 30, 47, 38, 73, 43 & 41

Step#1 Ho: $\sigma^2 \neq 100$

H₁: $\sigma^2 \neq 100$

Step#2 Level of significance: $\alpha = 0.05$

Step#3 Test Statistics

$$\chi^2 = \frac{(n-1)s^2}{\sigma_o^2}$$

Step#4 Critical region

$$\chi^2 > \chi_{0.025,9}^2 \text{ or } \chi^2 < \chi_{0.975,9}^2$$

$$\chi^2 > 19.023 \text{ or } \chi^2 < 2.7$$

If critical region satisfies reject H_0

Step#5 Calculation:

53, 46, 39, 39, 30, 47, 38, 73, 43 & 41

$$n = 10, s^2 = 135.433333, \sigma_o^2 = 100$$

$$\chi^2 = \frac{(n-1)s^2}{\sigma_o^2} = \frac{9 * 135.4333}{100} = 12.18899$$

12.1889 > 19.023 or 12.1889 < 2.7 → doesn't satisfied

Step#6 Conclusion: As Cr doesn't satisfied therefore there is no evidence to reject H_0 at 0.05 level of significance we conclude that variance is equal to 100.

Statistical Inference for Comparison Between Two Variances

Introduction

The **F-test** is used to compare the variances of two populations when the population variances are unknown. It helps determine whether there is a significant difference between the variability of two datasets. This method is widely used in fields such as finance, quality control, and research, where understanding consistency or risk is important. The test is based on the ratio of two sample variances and follows the F-distribution.

Point Estimate

The point estimate of the ratio of two population variances is:

$$\frac{\sigma_1^2}{\sigma_2^2} = \frac{s_1^2}{s_2^2}$$

This is simply the ratio of the two sample variances.

Confidence Interval for F ratio $\frac{\sigma_1^2}{\sigma_2^2}$

$$\frac{s_1^2}{s_2^2} * \frac{1}{F_{\frac{\alpha}{2}, v_1, v_2}} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} * F_{\frac{\alpha}{2}, v_2, v_1}$$

Where :

→ s_1^2 is sample 1 variance, s_2^2 is sample 2 variance

→ $v_1 = n_1 - 1$ is degree of freedom of sample 1, $v_2 = n_2 - 1$ is degree of freedom of sample 2.

Hypothesis Testing of Variance

There are 6 basic steps:

1. State the claim & its opposite
2. Level of significance (α)
3. Test statistic
4. Critical region
5. Calculation
6. Conclusion

Roadmap of Hypothesis testing for a single variance

Hypothesis	Test Statistics	Critical region
$H_1: \sigma_1^2 > \sigma_2^2$	$F = \frac{s_1^2}{s_2^2}$	$F > F_{\alpha, v_1, v_2}$
$H_1: \sigma_1^2 < \sigma_2^2$		$F > F_{\alpha, v_2, v_1}$
$H_1: \sigma_1^2 \neq \sigma_2^2$		$F > F_{\frac{\alpha}{2}, v_1, v_2} \text{ or } F > F_{\frac{\alpha}{2}, v_2, v_1}$

Exercise 9.5 elementary Statistics by Bluman

Question# 7. Ages of Hospital Patients The average age of hospital inpatients has gradually increased to 52.5 years. Studies of two major health care systems found the following information. At the 0.05 level of significance is there sufficient evidence to conclude a difference between the two variances?

	System 1	System 2
Sample size	60	60
Sample mean	49.8	50.2
Sample standard deviation	5.4	7.6

Source: *New York Times Almanac*.

Step#1 $H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \neq \sigma_2^2$

Step#2 Level of significance: $\alpha=0.05$

Step#3 Test Statistics

$$F = \frac{s_1^2}{s_2^2}$$

Step#4

Critical region

$$F > F_{\frac{\alpha}{2}, v_1, v_2} \text{ or } F > F_{\frac{\alpha}{2}, v_2, v_1}$$

$$F > F_{0.025, 59, 59} \text{ or } F > F_{0.025, 59, 59}$$

$$F > F_{0.025, 59, 59} \text{ or } F > F_{0.025, 59, 59}$$

$$F > 1.67$$

If critical region satisfies reject H_0

Step#5 Calculation:

	System 1	System 2
Sample size	60	60
Sample mean	49.8	50.2
Sample standard deviation	5.4	7.6

Source: *New York Times Almanac*.

$$s_1^2 = 5.4^2, s_2^2 = 7.6^2$$

$$F = \frac{s_1^2}{s_2^2} = \frac{5.4^2}{7.6^2} = 0.5048$$

$0.5048 > 1.67 \rightarrow$ doesn't satisfied

Step#6 Conclusion: As CR doesn't satisfied therefore there is no evidence to reject H_0 at 0.05 level of significance we conclude that both systems have same variation.

Question# 18. Elementary School Teachers' Salaries A researcher claims that the variation in the salaries of elementary school teachers is greater than the variation in the salaries of secondary school teachers. A sample of the salaries of 30 elementary school teachers has a variance of \$8324, and a sample of the salaries of 30 secondary school teachers has a variance of \$2862. At $\alpha=0.05$, can the researcher conclude that the variation in the elementary school teachers' salaries is greater than the variation in the secondary school teachers' salaries?

Step#1 $H_0: \sigma_1^2 \leq \sigma_2^2$

$H_1: \sigma_1^2 > \sigma_2^2$

Step#2 Level of significance: $\alpha=0.05$

Step#3 Test Statistics

$$F = \frac{s_1^2}{s_2^2}$$

Step#4

Critical region

$$F > F_{\alpha, v_1, v_2}$$

$$F > F_{0.05, 29, 29}$$

$$F > F_{0.05, 29, 29}$$

$$F > 1.85$$

If critical region satisfies reject H_0

Step#5 Calculation:

A sample of the salaries of 30 elementary school teachers has a variance of \$8324, and a sample of the salaries of 30 secondary school teachers has a variance of \$2862. At $\alpha=0.05$, can the researcher conclude that the variation in the elementary school teachers' salaries is greater than the variation in the secondary school teachers' salaries?

$$s_1^2 = 8324, s_2^2 = 2862$$

$$F = \frac{s_1^2}{s_2^2} = \frac{8324}{2862} = 2.9085$$

$$2.9085 > 1.85 \rightarrow \text{satisfied}$$

Step#6 Conclusion: As C_r satisfied therefore there is evidence to reject H_0 at 0.05 level of significance we conclude variation of elementary school teachers is greater than secondary school teachers